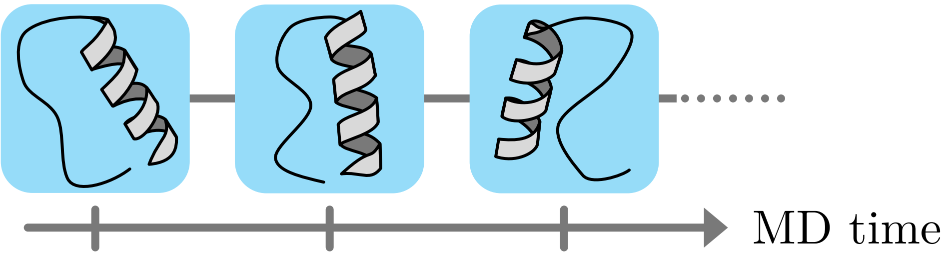
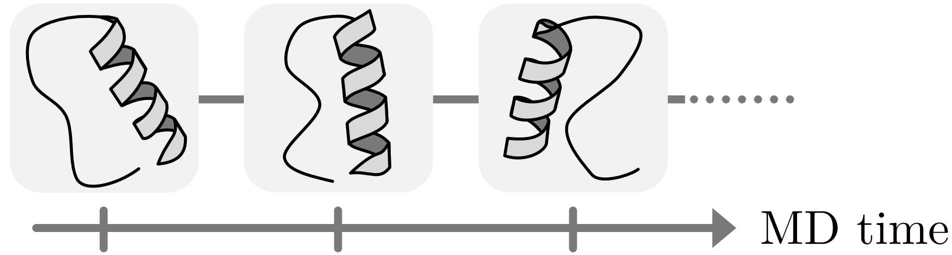
Type 1: MD simulation of the solution with leaving the solute flexible



$E(\psi)$ and solute structures ψ obtained

Type 2: MD simulation of the flexible solute at isolation



$\Delta_{\mathrm{A}\to\mathrm{B}}G_{\mathrm{iso}}$ and solute structures ψ obtained

Type 3: Calculation of the solvation free energy $\nu^{\mathrm{solv}}(\psi)$ at fixed structure of the solute

$$\sum_{\psi\in\mathrm{soln}} \mathcal{F}\big(\nu^{\mathrm{solv}}(\psi) - \mathcal{D}\big) = \sum_{\psi\in\mathrm{iso}} \mathcal{F}\big(-\nu^{\mathrm{solv}}(\psi) + \mathcal{D}\big),\ \mathcal{D} = \mu^{\mathrm{solv}} + k_{\mathrm{B}}T\log\frac{N_{\mathrm{soln}}}{N_{\mathrm{iso}}}$$

$$T\Delta_{\mathrm{A}\to\mathrm{B}}S^{\mathrm{conf}} = \Delta_{\mathrm{A}\to\mathrm{B}}\left\langle E(\psi) + \nu^{\mathrm{solv}}(\psi)\right\rangle_{\mathrm{soln}} - \Delta_{\mathrm{A}\to\mathrm{B}}G_{\mathrm{iso}} - \Delta_{\mathrm{A}\to\mathrm{B}}\mu^{\mathrm{solv}}$$